Evaluating and Graphing Exponential Functions
Answer Key

Algebra 1

Quick Answers

1. f(0) = 3, f(3) = 24
2. g(2) = 25, g(0) = 1
3. at x = -1 = $1 \cdot \frac{1}{2}$, at x = 0 = 1, at x = 1 = 2, at x = 2 = 4, at x = 3 = 8
4. at x = 0 = 100, at x = 3 = 12.5
5. x = 6
6. at x = 0 = 16, at x = 1 = 8, at x = 2 = 4, at x = 3 = 2, at x = 4 = 1
7. balance after 6 years = 1072.08
8. x = 4
9. f(4) = 162, f(-2) = $\frac{2}{9}$
10. at x = 0 = 4, at x = 1 = 8, at x = 2 = 16, at x = 3 = 32, x where y = 64 = 4, —
11. x = 5
12. (a) after 9 hours = 4000, (b) after 5 hours = 1587.40, (c) hours to reach 16000 = 15

What is verified

11 of 12 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1
HSF-IF.C – *problems 3, 6, 10*
HSF-LE.A.1 – *problems 1, 2, 4, 7, 9, 12*
HSF-LE.A.4 – *problems 5, 8, 11*
Difficulty 1–5 of 5 across 31 tagged checks.

Common wrong answers

9. *If they got* -18*: treated the negative exponent as a negative answer instead of a reciprocal*

1. $f(x) = 3 \cdot 2^x$. Find $f(0)$ and $f(3)$.

Solution (a): $2^0 = 1$, so $3 \cdot 1$. $\boxed{f(0) = 3}$

Solution (b): $2^3 = 8$, so $3 \cdot 8$. $\boxed{f(3) = 24}$

2. $g(x) = 5^x$. Find $g(2)$ and $g(0)$.

Solution (a): $5 \cdot 5$. $\boxed{g(2) = 25}$

Solution (b): Any nonzero base to the power 0 is 1. $\boxed{g(0) = 1}$

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3. $h(x) = 2^x$: complete the table for $x = -1, 0, 1, 2, 3$ and graph it.

Solution: A negative exponent is a reciprocal: $2^{-1} = \frac{1}{2}$. $\boxed{h(-1) = \frac{1}{2}}$

$2^0 = 1$. $\boxed{h(0) = 1}$

$2^1 = 2$. $\boxed{h(1) = 2}$

$2^2 = 4$. $\boxed{h(2) = 4}$

$2^3 = 8$. $\boxed{h(3) = 8}$

The plotted points $(-1, 0.5)$, $(0, 1)$, $(1, 2)$, $(2, 4)$, $(3, 8)$ rise ever more steeply and flatten towards the x-axis on the left without meeting it.

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4. $p(x) = 100 \cdot (0.5)^x$. Find $p(0)$ and $p(3)$.

Solution (a): $(0.5)^0 = 1$, so $100 \cdot 1$. $\boxed{p(0) = 100}$

Solution (b): $(0.5)^3 = 0.125$, so 100×0.125 — or halve three times: $100, 50, 25$. $\boxed{p(3) = 12.5}$

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5. Solve $2^x = 64$.

Solution: Write the right side as a power of the same base: $64 = 2^6$. Equal powers of the same base force equal exponents. $\boxed{x = 6}$

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6. $q(x) = 16 \cdot \left(\frac{1}{2}\right)^x$: complete the table for $x = 0$ to 4 and graph it.

Solution: At $x = 0$ the power is 1, so $16 \cdot 1$. $\boxed{q(0) = 16}$

Each step halves: $16 \div 2$. $\boxed{q(1) = 8}$

$8 \div 2$. $\boxed{q(2) = 4}$

$4 \div 2$. $\boxed{q(3) = 2}$

$2 \div 2$. $\boxed{q(4) = 1}$

The curve falls steeply at first and then flattens — decay, because the multiplier is between 0 and 1.

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7. \$800 growing 5% a year: find the balance after 6 years.

Solution: $1.05^6 = 1.340095640625$, so $800 \times 1.340095640625 = 1072.0765\ldots$, rounded to two decimal places. $\boxed{1072.08}$

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8. Solve $5 \cdot 3^x = 405$.

Solution: Divide both sides by 5 before touching the exponent: $3^x = 81$. Then $81 = 3^4$, so the exponents match. Check: $5 \cdot 3^4 = 5 \cdot 81 = 405$ ✓ $\boxed{x = 4}$

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9. $f(x) = 2 \cdot 3^x$. Find $f(4)$ and $f(-2)$.

Solution (a): $3^4 = 81$, so $2 \cdot 81$. $\boxed{f(4) = 162}$

Solution (b): $3^{-2} = \frac{1}{9}$, so $2 \cdot \frac{1}{9}$. The minus sign is in the exponent, not in front of the answer. $\boxed{f(-2) = \frac{2}{9}}$

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10. $y = 4 \cdot 2^x$: table, graph, solve $4 \cdot 2^x = 64$, and why the curve misses the axis.

Solution: $2^0 = 1$, so $4 \cdot 1$. $\boxed{4}$

$2^1 = 2$, so $4 \cdot 2$. $\boxed{8}$

$2^2 = 4$, so $4 \cdot 4$. $\boxed{16}$

$2^3 = 8$, so $4 \cdot 8$. $\boxed{32}$

Solving: divide by 4 to get $2^x = 16 = 2^4$, so the exponents match. $\boxed{x = 4}$

— **instructor-judged.** A correct answer says the curve never meets the horizontal axis because the output is 4 multiplied by a power of 2, and a power of 2 is always strictly positive however negative the input becomes; halving repeatedly shrinks the value towards zero without ever reaching it. Full credit for stating that the output is always positive and giving the never-reaches-zero reason; partial credit for saying only that the curve gets close to the axis.

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11. Solve $\left(\frac{1}{2}\right)^x = \frac{1}{32}$.

Solution: $\frac{1}{32}$ is $\left(\frac{1}{2}\right)^5$, since halving five times from 1 gives $\frac{1}{32}$. Same base, so the exponents match. $\boxed{x = 5}$

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12. Culture doubling every 3 hours from 500: count $= 500 \cdot 2^{t/3}$. (a) $t = 9$. (b) $t = 5$. (c) When does it reach 16,000?

Solution (a): $t/3 = 3$, so the count is $500 \cdot 2^3 = 500 \cdot 8$ — three doublings. $\boxed{4000}$

Solution (b): $t/3 = \frac{5}{3}$, and $2^{5/3} = 3.174802\ldots$, so $500 \times 3.174802 = 1587.401\ldots$, rounded to two decimal places. $\boxed{1587.40}$

Solution (c): Divide by 500: $2^{t/3} = 32 = 2^5$, so $\frac{t}{3} = 5$. Check: $500 \cdot 2^5 = 16000$ ✓ $\boxed{t = 15}$